

Judge	Score
Judge 1	Conceptual Design: I thought your decomposition of requirements was one of the better that I have read in grading reports. I also think the sensitivity discussion is well done. You probably could have added some other plot that could highlight that the maximum competition score does not align with the maximum mission 2 or 3 score. Three of your landing gear configurations have the exact same text. Appears to be an editing error. Preliminary Design: Overall good. Could be improved by documenting some of the key equations used in your code within the text for reference. Drawing Package: As all the rules and requirements are in English units, in this case using them in the report would be preferred. Test Plan: You could have added some additional details on how the tests were setup and performed for some of these tests. Just looking for a little more detail in this section Performance Results: Again could use more details. Why was 2 ply so much worse than 1 or 3 ply, etc. Again could use some additional discussion in this section. Bibliography Results: Should also reference a source for the competition rules.

Judge 2	Overall good job. In different sections, some subcomponents were discussed but not others. All subcomponents should be evaluated at each step of the design process. This led to the loss of some points.
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- General - No need to show a detailed table of contents. Fit it on one page and take advantage of every page you're given (60) - 1 Executive Summary - Great executive summary - 2 Management Summary - Missing specific dates of key deliverables on schedule - Should show more schedule detail of subtasks and subcomponents - 3 Conceptual Design - I like the discussion in section 3.2.1 - In table 3.2, is the given CL for cruise, max, takeoff? You must be explicit - Inconsistent units. In US aviation, we use English units. Don't have a problem with Metric here per se, but don't mix (table 3.2) - Great discussion and presentation of data throughout Section 3.3. I would like to see some reference to a specific score you think you can obtain based on your selected design values. Make assumptions for competition max or present un-normalized scores - 4 Preliminary Design - Is the crew included in table 4.1? - Should present more than just longitudinal static stability - I recommend converting your dynamic stability analysis into some aspect of time to half/double/tenth/10x as those are generally more interpret-able. But good analysis nonetheless - Mission performance should include specific mission parameters like number of laps, payload weight, number of passengers, battery capacity, etc. - 5 Detail Design - Would like to see more discussion on expected flight loads - There should be more discussion on propulsion and avionics in the detail design section - Did you consider the crew in your weight and balance? - There are 4 missions in this competition. Do not leave out the ground mission in your final aircraft performance - The battery capacity is a key scoring parameters. It should be included in your final aircraft performance - Drawing Package - Take advantage of the exploded view to show more detail on the structural layout of the aircraft... key components, interfaces, etc. Would like to see the aircraft laid out as it is assembled, the wing should at least be facing the same direction as the fuselage and tail. - Should show dimensions on parking configuration as that is a key dimensional requirement - You need more detail about the mission configurations i.e. show gurney outside the aircraft with dimensions, etc. - 6 Manufacturing Plan - Would like to see more discussion, perhaps even decision matrices, in section 6.1.6 - 7 Testing Plan - Figure 8.1 seems unnecessary. Just share the total thrust value in the given propulsion configuration - Bibliography - What about the AIAA rules? - Good job with references and their use throughout the text

Executive Summary: How will you meet the parking requirement? Management Summary: How many students do you have on your team? How is communication assured between sub-teams? How are decisions made? Schedule should include specific dates for key deliverables. Schedule: would like to see more detail on design and testing phases. What are the various subsystems you are designing, constructing and testing? Conceptual Design: 3.2.1: nice discussion of conflicting design considerations. I would like to have seen some plots of the data you discuss that produced table 3.3. I would like more discussion of how you generated Figure 3.2, but your analysis of what it means is thorough. Typo at end of 3.4.2 says you chose the high-wing. Preliminary Design: Section 4.4.2: I suspect that your "...past WUDBF projects" are not actually referred to in Reference 1 as indicated. The software packages you use in your analysis need to be properly referenced, and included in your bibliography. Would like to see more details (and/or simple example) of how your Km_sim performs mission simulation calculations. How did you determine that 2.5G is an appropriate load factor for turns? Many RC aircraft see much higher loads. You provide lift and drag coefficients, but what do you expect the overall aircraft lift and drag to be? What weight of M2 cabinet are your simulations based on? How many M3 passengers are your simulations based on? Detail Design: You discuss load paths, but never talk about max loads expected. Would like to see discussion of your propulsion and avionics systems, including battery capacities selected for the various missions. 5.3.5 I am unclear on your landing gear attachment – does it bolt to the wingbox as stated, or just to the wing rotation bearing? Would like a bit more detail on your nose gear. Note: a common coordinate system used in industry for W&B calcs orients +X aft, +Y right, +Z up, but it is possible to work with any coordinate system as long as it is defined (as yours is). How do your CG locations compare to your estimated location that you used to determine your stability margins? Based on dims in your drawings, your aircraft CG is over 40% of your MAC for M1, and over 50% of your MAC for M3 – center of lift if typically forward of 30% MAC location, which would make your aircraft very unstable. Have you confirmed that your CG locations are accurate? 5.3.6 What is the design of your medical cabinet, and what is its max weight? 5.5.1 you compare stress resulting from motor thrust loading to fiberglass strength. How does it compare to the layup strength (i.e. fiber direction and matrix selection have large impact on strength of a part). Also, you did not reference your plywood strength - where did it come from? Did you also evaluate your motor mount for loads associated with flight (i.e. G loads on the motor mass in combination with and transverse to the thrust loading)? In all your analyses you are looking at von Mises stresses – in some cases these are compressive, so compression/crippling/buckling need to be evaluated. 5.5.2/5.5.3 The 2.5 load factor you applied may not be sufficient to show that your structure is adequate. The 2.5G (effective) wingtip load test is not intended to be the ultimate design load for your aircraft – you are likely to see much higher loads in flight. Your CF spar does not have as large of a safety margin as you think it does, considering higher actual load factors and potential compressive bending failure. Table 5.3: what is your cabinet weight for M2, what is your passenger count & battery capacity for M3? Drawing Package: Should include detail breakouts for your Medical supply cabinet, gurney, and passenger/crew restraints. I only see one battery in your drawings – is this the propulsion battery or the receiver battery? Where is the other one located? Need a separate systems layout drawing that shows where all your servos/batteries/receiver/etc are located. Where do the wires run from your wing servos to the receiver? Manufacturing Plan: Would like to see more details of your decision-making process as it relates to choosing materials and manufacturing methods. Test Plan: Would like to see a bit more detail on your fiberglass structural test. Would be good to include sketches/descriptions of the test specimens (including fiber orientation) and test setup. Performance Results: Would be good to include data from your test trials for GM test. One of the goals of your flight testing was to verify take-off distance. What did you learn regarding this during the flights you had? Bibliography Results: References should include your software packages (MATLAB, Simulink, XFLR5, Ansys, Xfoil), and (at minimum) the rules document.